Stretch meeting

Willy Ware report — Aug 20, 1956

physical research

IBM (Theory)
A. Eschenfelder — ferro. materials Theory
G. Tucker — solid State materials (transistors) Theory
diff transistors, etc.

(Production) component research,
Mgr J.P. Little
asst Mgr
Pough: in charge
Palmer - Research in Pough

magnetics
mechanical
analysis testing
process tech.

semiconductors Bob Swanson

mag films - back up effort { less drive current 5000 Å thick better fabricability,
ferrites 0.1-0.2 μsec flip times

IBM
Present Production { 8000 transistors/mo, "slow" NPN, PNP
100000 cycle circuits 20-30% good yields

plans to expand 2 fold by Jan '57
13 (700 machines on production at once)

{ 4,000,000 good cores per mo. 80% yield
(10 μs stores)

printed circuits — very well in hand, — both sides of boards

have 500 or so of these

are planning 4 types { 2 MP drivers high voltage heavy current PNP NPN
10 MP switching, logical PNP NPN

factory is getting ready for pilot production,

2 or 3 methods of making drift transistors are being studied,

Wm. Lawrence - Stores
$\frac{1}{4}" \times \frac{1}{2}"$ down to microscopic size

(20 mμsec 0 - 4 amps current drive rise time.)

Hanley, - transistors,
- have been working on SAC or Sage System,
Philco SBT's,

adder: 0.5 μs carry time per stage
→ 1 transistor to drive 4 others,
question of tolerances in design,

IBM #738. 32000 word store for 704

(Sept 57 contr. start) Time Schedule: - construction 18 mos from now
need 100,000 transistors
Pub's = 0.6 for 42 mos.
= 1.0 for 54 mos.

Initial Planning - W. Buchholz in charge, Greenwald

Suggestions:

Input output processor – central computer

→ – how do they communicate w.r.t. each other? keeping out of each others way.
could be difficult to program

- 60 bits not 48 bits only for logical rd. ?
- multiple tape operation – (is provided)
max. efficiency for whole system? compiler
worth money?
→ skip instructions which occur when toggles go on.

$4,3 M has no overhead, nor R+D . just the construction + maint. cost.

Philco transp $6.00 each.

1. processor,
2. look ahead feature
} two newest features

→ noise problems will probably be great.
5 volt swings (704 is 40 volts or so)